

Introduction to Computer Science

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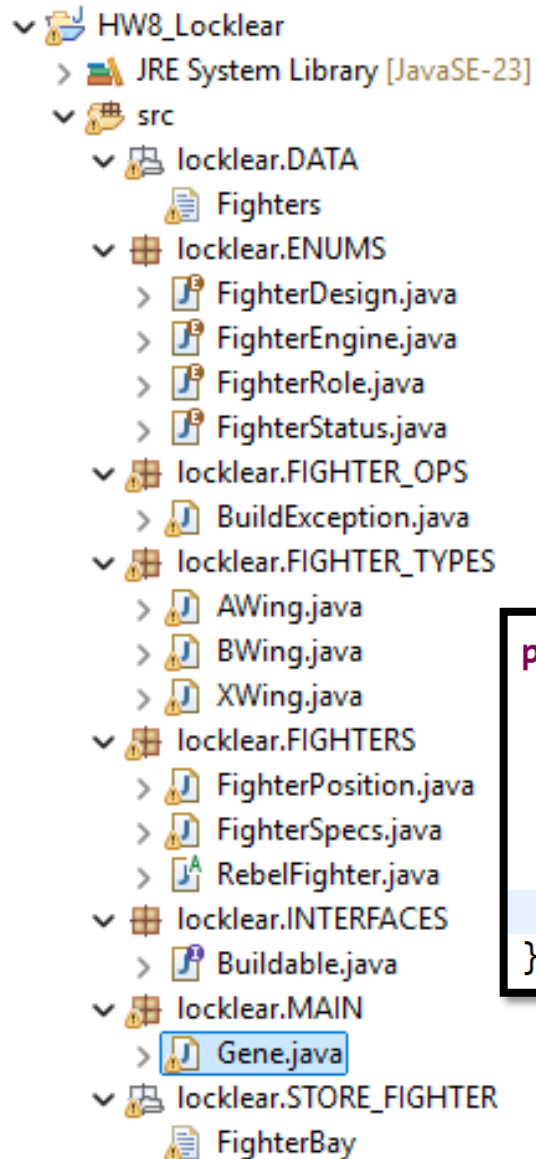
Homework 8

Java OOP Mastery

Professor HG Locklear

Program Structure

Create your Program Structure and main method
EXACTLY as shown replacing my name with
yours where appropriate



```
public static void main(String[] args) throws IOException, BuildException {  
    Buildable.buildFighters("Fighters");  
    Buildable.displaySpecifiedFighter("GAB17");  
    Buildable.storeSpecifiedFighter("IIA15", "FighterBay");  
    Buildable.rebelFighters.get("AIB06").getfPos().displayLocation();  
    System.out.println(Buildable.rebelFighters.get("CCC73").getfDesign().getDescription());  
    AWing A = new AWing("ABC1", FighterStatus.Unserviceable, new FighterPosition(0,0));  
}
```

Special Instructions

- During the Galactic Civil War, the Rebel Alliance relied on a diverse fleet of starfighters to counter the numerically superior Imperial Navy.
 - The **A-Wing** served as the fleet’s lightning-fast interceptor, designed for rapid response, reconnaissance, and hit-and-run attacks against vulnerable Imperial assets.
 - The **B-Wing** was the Alliance’s heavy assault starfighter, bristling with weaponry and optimized for punching through capital ship defenses during fleet engagements.
 - The **X-Wing**, often called the backbone of the Rebel Starfighter Corps, was a versatile space superiority fighter capable of both dogfighting and precision strikes.
- Together, these starfighters provided the Alliance with the flexibility to engage the Empire in space battles across a range of mission profiles—from high-speed interception to heavy assault and all-purpose combat operations.

Fighter	Length	Width	Height	Mass	Engine	Role
A-Wing	6.9m	4.47m	2.47m	12 metric tons	J77	Interceptor
B-Wing	16.24m	8.54m	2.44m	19 metric tons	JL4	Bomber
X-Wing	12.5m	11m	4.4m	10 metric tons	R200	Fighter

The **Thrust-Mass** ratio of any fighter is the **Engine Thrust / Fighter Mass**

Enumerations

<ENUM>
FighterDesign

AWING("Most often deployed to disrupt enemy formations, chase down TIE Interceptors, or carry out quick strikes against key targets.")

BWING("Most often used to punch through Imperial capital ship defenses or disable critical systems " with ion weaponry before bomber follow-up")

XWING("Most often used as a space superiority fighter and as an escort for AWing and BWing attack craft.")

-String description

-FighterDesign(String description)

<ENUM>
FighterEngine

AWING("J77",120)
BWING("JL4",100)
XWING("R200",115)

-String engineType
-double thrust

-FighterEngine(String engineType, double thrust)

<ENUM>
FighterRole

Interceptor, Bomber, Fighter

<ENUM>
FighterStatus

Serviceable, Unserviceable

BuildException Class

Exception

<https://docs.oracle.com/javase/8/docs/api/java/lang/Exception.html>

BuildException

-String fighterID

BuildException(String fighterID)

The **BuildException** is thrown by any RebelFighter subclass at instantiation when the length of the **fID** attribute is **less than five characters/digits**. The exception message is **formatted as shown below**.

Exception in thread "main" [locklear.FIGHTER OPS.BuildException](#): Fighter ID Must Have 5 Letters/Digits: ABC1 is INVALID

FighterPosition Class

FighterPosition

-int x

-int y

FighterPosition(int x, int y)

+displayLocation(): void

+distanceTo(FighterPosition pos): double

<<Serializable>>

FighterPosition Method Specification Chart			
Method	Input	Processing	Output
displayLocation	None	Print to the Console a String representation of the FighterPosition object formatted as shown (See Slide 12)	Console
distanceTo	pos	Returns the 2D Euclidean distance between the two FighterPosition objects	double

FighterSpecs Class

FighterPosition
-TreeMap<String, Double> dimensions -FighterEngine engine
+displayFighterSpecs(): void +displayFighterEngine(): void

<<Serializable>>



The **dimensions** attribute contains the attribute values for **length**, **width**, **height**, and **mass** with their key being the name of the attribute (i.e., Length)

FighterSpecs Method Specification Chart			
Method	Input	Processing	Output
displayFighterSpecs	None	Prints to the Console the length , width , height , and mass of the RebelFighter object contained in the dimensions attribute in the format shown (See Slide 12)	Console
displayFighterEngine	None	Prints to the Console a String representation of the engine attribute in the format shown (See Slide 12)	Console

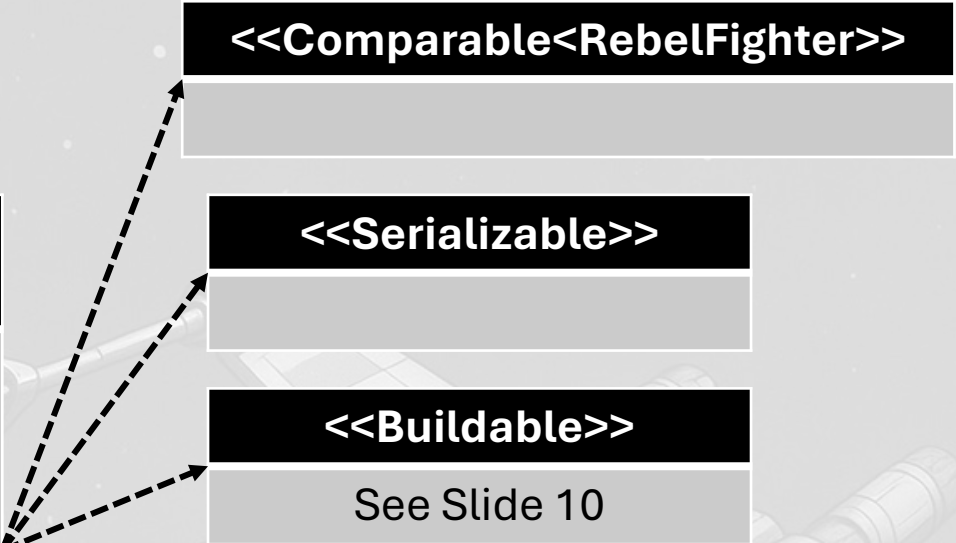
RebelFighter Class

<Abstract>
RebelFighter

-String fID
-FighterRole fRole
-FighterSpecs fSpecs
-FighterDesign fDesign
-FighterStatus fStatus
-FighterPosition fPos

RebelFighter(String fID, FighterRole fRole, FighterSpecs fSpecs, FighterDesign fDesign, FighterStatus fStatus, FighterPosition fPos)

+displayFighter(): void



FighterPosition Method Specification Chart			
Method	Input	Processing	Output
displayFighter	None	Print to the Console a String representation of the RebelFighter object formatted in any format you like	Console

RebelFighter Subclasses

<Abstract>
RebelFighter

See Slide 7

Subclasses **must use** their superclass constructor

AWing

AWing(String fID, FighterStatus fStatus, FighterPosition fPos)

+compareTo(RebelFighter o): int
+showFighterDetails(): void

BWing

BWing(String fID, FighterStatus fStatus, FighterPosition fPos)

+compareTo(RebelFighter o): int
+showFighterDetails(): void

XWing

XWing(String fID, FighterStatus fStatus, FighterPosition fPos)

+compareTo(RebelFighter o): int
+showFighterDetails(): void

Subclass Method Specification Chart			
Method	Input	Processing	Output
compareTo	o	Compares this RebelFighter object to o based on their fID attribute	int
showFighterDetails	None	Prints to the Console a String representation of the RebelFighter object formatted as shown (See Slide 12)	Console

Buildable Interface

<<Buildable>>

```
+static String dirData = relative filepath to DATA
+static String dirStore = relative filepath to STORE_FIGHTER
+static TreeMap<String, RebelFighter> rebelFighters = new TreeMap<>()
+abstract showFighterDetails( ): void
+static buildFighters(String filename): void
+static displaySpecifiedFighter(String fID): void
+static storeSpecifiedFighter(String fighterID, String filename): void
```

Buildable Interface Methods

Buildable Method Specification Chart			
Method	Input	Processing	Output
showFighterDetails	None	See RebelFighter subclasses	Console
buildFighters	filename	Reads the file filename and utilizes the data to construct the specified RebelFighter objects as defined on each row and stores each RebelFighter object in the rebelFighters TreeMap	Container
displaySpecifiedFighter	fID	Calls the showFighterDetails method of the specified RebelFighter object fID contained in the rebelFighters TreeMap	Console
storeSpecifiedFighter	fighterID filename	Serializes the specified RebelFighter object fighterID from the rebelFighters TreeMap in the file filename and prints to the console “[fID] has been stored”	File Console

Slide 12

showFighterDetails

X-WING

Fighter ID: GAB17	Fighter Status: Unserviceable	Fighter Role: Fighter
Length: 12.5	Width: 11.0	Height: 4.4
Mass: 10.0	Engine Type: R200	Engine Thrust: 115.0
Thrust-Mass Ratio: 11.50		

displayLocation

Location: 14 14

displayFighterSpecs

Length: 12.5	Width: 11.0	Height: 4.4	Mass: 10.0
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displayFighterEngine

Engine Type: R200	Engine Thrust: 115.0	Thrust-Mass Ratio: 11.50
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Fighters File

Fighters X

```
1AIB06 A-Wing Unserviceable 14 2
2CCC73 X-Wing Serviceable 17 6
3IDD01 A-Wing Serviceable 6 3
4IIA15 A-Wing Unserviceable 19 28
5DAB46 A-Wing Unserviceable 18 17
6DIG17 X-Wing Unserviceable 2 13
7IID13 B-Wing Serviceable 11 11
8GAB17 X-Wing Unserviceable 16 6
9DGE43 A-Wing Unserviceable 11 17
10EEI74 A-Wing Serviceable 17 13
```

Each row of the file **represents an individual** RebelFighter object specified by

- Fighter ID
- Fighter Type
- Fighter Status
- Position X Coordinate
- Position Y Coordinate

This file is provided

Expected Output

```
***X-WING***
Fighter ID: GAB17      Fighter Status: Unserviceable  Fighter Role: Fighter
      Length: 12.5    Width: 11.0    Height: 4.4    Mass: 10.0
      Engine Type: R200    Engine Thrust: 115.0    Thrust-Mass Ratio: 11.50
IIA15 has been stored
Location: 14 2
Most often used as a space superiority fighter and as an escort for AWing and BWing attack craft.
Exception in thread "main" locklear.FIGHTER_OPS.BuildException: Fighter ID Must Have 5 Letters/Digits: ABC1 is INVALID
    at locklear.FIGHTERS.RebelFighter.<init>(RebelFighter.java:25)
    at locklear.FIGHTER_TYPES.AWing.<init>(AWing.java:14)
    at locklear.MAIN.Gene.main(Gene.java:42)
```